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Modular Routing Design for Chiplet-based Systems

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***Abstract*—System-on-Chip (SoC) complexity and the increasing costs of silicon motivate the breaking of an SoC into smaller "chiplets." A chiplet-based SoC design process has the promise to enable fast SoC construction by using advanced packaging technologies to tightly integrate multiple disparate chips (e.g., CPU, GPU, memory, FPGA). However, when assembling chiplets into a single SoC, correctness validation becomes a significant challenge. In particular, the network-on-chip (NoC) used within the individual chiplets and across chiplets to tie them together can easily have deadlocks, especially if each chip is designed in isolation.**

We introduce a simple, modular, yet elegant methodology for ensuring deadlock-free routing in multi-chiplet systems. As an example, we focus on future systems combining chiplets on an active silicon interposer. To maximize modularity, each individual chiplet is free to implement its own NoC topology and local routing algorithm, and the interposer can implement its own independent topology and routing. Our methodology imposes a few simple turn restrictions applied only to traffic as it flows into or out of the chiplets from the interposer, and we provide a way to determine these restrictions. The end result is an overall approach that enables highly-modular, chiplet-based SoC construction while eliminating deadlocks with high performance.

***Keywords*-chiplet; deadlock-avoidance; routing**

I. Introduction

The rising costs of large Systems-on-Chip (SoCs) in increasingly complex process technologies are a motivation in the design of SoCs based on "chiplets." This concept breaks a conventional monolithic SoC into several smaller chiplets, each of which can be cheaper to develop, easier to reuse across multiple products, and implemented with the most appropriate process technology. The chiplet approach is being actively worked on in academia [1]–[4], industry [5]–[11], and government agencies [12].

There are many research and engineering challenges associated with chiplet-based architectures; we focus on one specific but critical problem. Adhering to a modular design approach, each individual chiplet should be designed and verified without any knowledge of the full system. When constructing a SoC from multiple chiplets, even though each individual component is properly verified, the fully-integrated system may still have correctness issues. The interconnection network is particularly vulnerable to this. Each individual chiplet may contain its own local network-on-chip (NoC) that is locally deadlock free and operates properly for *intra-chiplet* traffic. However, connecting several NoCs together can introduce new resource cycles that cause cyclic dependencies *across* the chiplets.

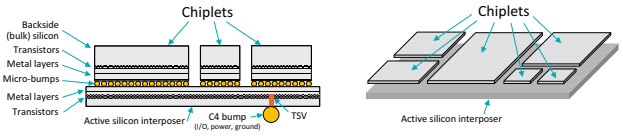

Figure 1: Example organization of a SoC implemented as multiple chiplets stacked on an active interposer.

While relatively simple chiplet-based architectures have been announced [13] or are even already available [14], this work looks further ahead to emerging architectures based on stacking multiple chiplets on active silicon interposers (although we also explain how to apply our methodology to nearer-term integration approaches such as passive interposers). We first provide some background on chiplets and describe the modularity challenges associated with existing multi-chiplet architectures. We then introduce a new chiplet-based routing methodology that enables each chiplet to be independently designed without knowledge of other chiplets or the interposer's NoC details, which is a key attribute not supported by prior art. Our composable routing approach leverages a simple-yet-powerful insight: from an individual chiplet's perspective, the rest of the system can be abstracted away into a single virtual node. Turn restrictions are carefully and easily applied to *only* the boundary routers that connect the chiplet to the virtual node, leading to tractable analysis and optimization at the granularity of individual chiplets.

II. Chiplet-based Systems

The slowing of Moore's Law and Denard Scaling have driven leading-edge process technologies to become increasingly complex and expensive. To offset the slowdown of scaling, many chips are getting bigger to continue generational improvements in functionality and performance; a recent example is the NVidia "Volta" GPU that uses a 815mm^2 chip [15]. Recently, industry and government are pursuing and advocating SoC designs based on the concept of "chiplets", where a large expensive SoC can be decomposed into multiple smaller, higher-yielding, and cost-effective chiplets that are then reassembled using advanced packaging technologies. These include AMD's exascale APU vision [6], [10], NVidia's MCM-GPU [11], TSMC's CoWoS (chip-on-wafer-on-substrate) services, Marvell's MoChi^{TM} (Modular Chips) architecture [7], [8], and DARPA's CHIPS program [12]. A chiplet approach also enables SoCs combining silicon from different companies, such as the recently-announced Intel Core processor with AMD Radeon Graphics technology [13]. The computer architecture research literature also reflects these trends with studies involving chiplet-like architectures using passive silicon interposers [16], passive

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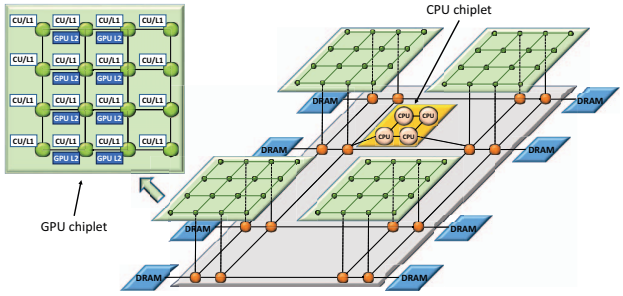

Figure 2: Baseline architecture.

interposers with microfluidic cooling [17], active silicon interposers [2], [18], and photonic chiplets [3], [4].

A. Active-interposer Chiplet SoCs

While current multi-chiplet architectures utilize passive integration technologies such as silicon interposers [6] and multi-chip modules [11], [14], this paper looks forward to chiplet-based SoCs based on emerging *active* silicon interposers, as shown in Fig. 1 (we do explore other packaging technologies in Section VI). Despite the near-term commercial focus of passive substrates (only wires but not logic) [19]–[21], there is increasing academic [2], [4], [22], industrial and government research [23]–[28] focused on active interposers. A working active interposer prototype with a 3D NoC has already been demonstrated [29].

Many common SoC functions can be moved to an active interposer, such as external memory interfaces, inter-chiplet connectivity (i.e., NoC), external IO, and system management and debug (e.g., reset, JTAG). This allows individual chiplets to be simpler (reduces design time) and smaller (improves yield/cost). If the chiplets are implemented in a more expensive technology node (e.g., 14nm) and the interposer is implemented in a more mature and less expensive process (e.g., 28nm, 20nm), there is an additional cost benefit from moving logic off the more expensive chiplets to the interposer [2]. A recent analysis concludes that active interposers can be cost-effective for large SoCs, even compared to passive silicon interposers [30].

B. Baseline Assumptions

While our proposed approach applies to a wide variety of possible chiplet-based SoCs, we focus on a specific architecture as a working example. We consider a multi-chiplet heterogeneous computing system (an "APU"), consisting of both CPU and GPU components. Fig. 2 shows the baseline system, which is optimized for GPU compute. There are four GPU chiplets, each providing 16 GPU SIMD compute units (CUs), and a central CPU chiplet to support the CPU phases of GPGPU workloads. These five chiplets are stacked on an active interposer that implements its own NoC to interconnect the chiplets and other common system functionality.

Our baseline configuration uses mesh topologies for the chiplet and interposer NoC sub-networks. Each GPU chiplet's 16 CUs are arranged in a 4×4 mesh, and the interposer layer also has a 4×4 mesh connecting the chiplets. All NoC components use static routing implemented with routing tables, as is typical for current commercial systems (e.g., HyperTransport [31] or QuickPath Interconnect (QPI) [32]). Each chiplet's local mesh and the interposer mesh use X-Y routing. Additional details such as NoC router configuration (e.g., buffer sizes, pipeline depth) are provided in Section V-A. Our baseline provides an APU with a total of 64 CUs of GPU compute, 4 CPU cores, and 8 external memory channels, while maintaining a relatively simple structure to aid in our explanations, evaluations, and analyses.

III. Motivation

We now describe how deadlock-free chiplets can induce deadlocks when connected together, and then we discuss the limitations of prior works including: global system-level knowledge of the SoC, high costs of required virtual channels, and restrictions on local routing algorithms. This section ends with a summary of what "modular" chiplet design means for this paper and how the different approaches measure up.

A. Networks on Chip

An NoC provides a uniform interface to connect different system components. Rather than forcing the system designer to implement specific interfaces between every pair of communicating blocks, and worse to validate the correct behavior of each of these interfaces, the NoC approach enables a far more modular and scalable design methodology that is a natural fit for tying together different chiplets.

Routing can significantly impact network performance, reliability, and functionality [33]. Improperly designed routing algorithms can cause resource dependencies in the network, leading to deadlocks that can be fatal to the system. We develop a modular, yet deadlock-free approach to routing in chiplet-based systems with various topologies.

B. Chiplet Composability Challenges

For multi-chip SoCs, chiplets could come from different vendors [13], and even when supplied by a single vendor they may be *independently designed* by different teams. As chiplets may be deployed in multiple products, including future products not even defined at chiplet-design time, global SoC routing information may not be available. Thus, designing the chiplets for use in *any* conceivable SoC or topology becomes extremely challenging, because while each individual chiplet's NoC may be deadlock-free, they can still be connected together in a manner that introduces deadlocks in the final SoC. Fig. 3a shows an example where two 4×4 mesh chiplets are connected through additional links. Although each individual chiplet uses deadlock-free X-Y routing, there still exists channel dependencies that can lead to deadlocks. Fig. 3b shows a two-chiplet interposer-based system with a few potential dependency loops highlighted.

Most existing deadlock-free routing algorithms assume that complete system-level information is available, which does not necessarily hold in chiplet-based systems. Therefore, these approaches are not amenable to routing for *modular, independently-designed* chiplets that may be reused in multiple SoC designs and topologies. We address this issue and propose a composable routing algorithm for the modular design of future SoCs.

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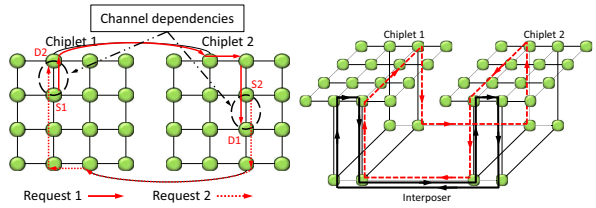

(a) 2D network. (b) chiplet-based system.

Figure 3: Deadlock scenarios. a) When two 2D-mesh networks are connected. b) Possible dependency loops in a two-chiplet interposer-based system

C. Deadlock Avoidance

Deadlock is avoided by preventing cycles in an NoC's resource dependency graph. There are two main techniques for avoiding cyclic dependencies: (1) virtual channel (VC) approaches [34], and (2) turn models [35], [36]. Turn models do not rely on additional virtual channels to prevent deadlock. Instead, they impose turn restrictions on certain paths to prevent cycles from forming. In this work, we leverage turn restrictions to ensure deadlock-freedom for a multi-chiplet, interposer-based NoC, but we introduce a routing methodology that only requires a few selected turn restrictions at the "boundaries" between the chiplets and the interposer. We now discuss related works and explain their limitations in the context of SoCs based on reusable, modular chiplets.

VC-based Approaches: VC-based approaches split a physical channel into multiple virtual channels in a time-multiplexed manner. Each VC is independently managed and has dedicated (per-VC) flit buffers in each NoC router. Circular dependencies are removed by assigning different network flows to disjoint VCs. Note that VCs for routing deadlock freedom are in addition to virtual networks required for protocol-level deadlock avoidance. Therefore, for a heterogeneous system requiring a complex coherence protocol, the number of VCs required may be quite large (impacting NoC router area, power, etc.). For composable chiplet-based systems, the number of VCs must be pre-configured to support the largest conceivable system, and all individual chiplets in the system *must* implement this for the largest number of VCs, leading to over-provisioning for smaller systems and higher costs for the individual chiplets.

Increasing the number of VCs has a direct impact on the area and power of the NoC routers as each VC has its own input buffers, and the arbitration logic scales with the number of VCs. From the perspective of chiplet-based systems, an individual chiplet's NoC can be designed with a different number of VCs to guarantee deadlock-freedom depending on its local topology and routing scheme; this makes it extremely complicated to design and verify the VC allocation/arbitration logic when integrating multiple such networks. To eliminate deadlocks using VCs, designers need to know the full system details ahead of time, over-provision VCs, and/or constrain the allowable per-chiplet and/or interposer NoCs. For these reasons, we seek alternatives to VC-based approaches to resolve deadlocks in multi-chiplet, interposer-based SoCs.

Flat Networks: One approach is to treat the entire system as a flat network and apply a unified global routing algorithm. Many topology-agnostic routing algorithms have been proposed in this context. The first such algorithm was up*/down* [37], which uses a breadth-first search (BFS) spanning tree formed from a root node. Links pointing toward the root are uplinks, while the rest are downlinks. Channel dependencies are avoided by forbidding messages to turn from a downlink to an uplink. Up*/down* routing requires analysis and programming of all routing tables in a global fashion, which does not allow the individual chips to make use of (better) local routing decisions. This also severely reduces the modularity and composability of the system. We also find that up*/down* routing leads to unbalanced traffic as links near the root node tend to be more congested than those near the leaf nodes.

Segment-based Routing partitions the network into subnets, and subnets into segments, and places bidirectional turn restrictions within each segment [38]. For a *starting segment* that forms a cycle, turn restrictions can be placed on any router except the starting one; for *regular segments*, cycles are broken by placing bidirectional turn restrictions on any router; for a *unitary segment* that consists of only one link, no traffic is allowed to cross the link (thus, on one side of the link a bidirectional turn restriction must be placed between this link and every other adjacent link). Optimization is possible because turn restrictions can be placed freely within a segment independent of other segments.

Nue [39] is a destination-based oblivious routing implementation for InfiniBand. Based on a complete channel dependency graph (CDG) of the network, Nue constructs a spanning tree that guarantees deadlock freedom and connectivity. Then it uses Dijkstra's algorithm to compute shortest paths from one source node to all other nodes in the complete CDG while maintaining the cycle-free constraint. Nue does not rely on VCs to provide deadlock freedom, although additional VCs do improve load balancing and performance.

All of these flat-routing approaches require global information of the target SoC to construct the CDG upon which deadlock freedom can be achieved. Full analysis of the CDG can be prohibitively expensive [40]. As discussed earlier, full-SoC configuration and topology information are not expected to be available for chiplet-based systems (e.g., chiplets may be reused for future yet-to-be-specified SoCs). While we provide experimental comparisons against several of these approaches, we emphasize that *none of these prior approaches satisfy the objective of enabling truly modular and reusable chiplets with independently-optimized chiplet-local NoCs*.

Hierarchical Approaches: Another approach is to break down an NoC topology into several hierarchical layers. In hierarchical routing, designers are free to choose any existing routing algorithm for a single level; and each node only knows about the local nodes within its level. A message destined to another level is first forwarded to a *boundary* router, which is connected to another level of hierarchy. From the source boundary router, the message is directed toward

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its destination through other boundary routers. An advantage of hierarchical routing is that each level of the local network is analyzed independently, and a locally-optimal routing algorithm can be applied. However, as we discussed earlier, when combining individual networks together, the global network could still have deadlocks. As a result, care must be taken to avoid global deadlocks, which usually results in *case-by-case analysis* of all possible global routing paths [41], [42]. Previous works proposed hierarchical NoCs based on regular topologies such as bus, ring, mesh, and tree [42]–[44]. However, chiplets from different manufacturers might not be designed with regular NoCs, and the integrated SoC system might not be symmetric. Consequently, system-level deadlock avoidance can still require great effort and is error-prone.

D. Comparison of Modularity

While there may be many possible definitions of "modularity," we focus on the key attributes listed in Table I.

Independently Designed Chiplets: Architects of an individual chiplet should be able to design and optimize their local NoC with little knowledge of the rest of the overall SoC(s). VC-based approaches require the chiplet architect to either have information for the overall SoC organization (chiplet design no longer independent) or to over-provision the number of VCs to support all possible SoCs in which the chiplet may be used. Flat and hierarchical NoCs also typically require full SoC information to analyze and ensure deadlock freedom.

Enables Local Optimization: A modular design approach should allow a chiplet architect to locally optimize a chiplet's NoC independent of the final SoC organization. Flat approaches require global SoC information, and therefore intra-chiplet optimizations impacting the local topology, routing algorithms, load balancing, etc., cannot be performed in isolation. Hierarchical NoCs do enable some degree of local chiplet-level optimization, although this still may be constrained by the global analysis of the full CDG to eliminate deadlocks. VC-based approaches as well as the Composable scheme proposed in this paper effectively allow arbitrary chiplet-level NoC organizations and optimizations.

Global CDG Not Required: Flat and Hierarchical NoCs are not modular because constructing the dependency graph requires all channels' connectivity, and route assignment cannot be performed until all chiplet networks are finalized. It is also extremely difficult to optimize the local NoC without impacting global routing decisions, as modifying the local network changes the global CDG. VC-based approaches offer more flexibility in local optimization and do not require the global CDG; they require some global information in order to assign VCs. Our Composable approach requires some limited information about a chiplet to be shared with the SoC integrator (but far less than the full set of channel dependencies), and *no* dependency information to be shared between independent chiplet designs.

Future-proof Chiplets: A chiplet could get integrated in a future SoC that has yet to be designed. As the Flat and Hierarchical NoC approaches require the global CDG, chiplets would be difficult to reuse as local NoC designs and optimizations may already be fixed. VC-based approaches are perhaps better off, but over-provisioning of VCs for yet-to-be-considered SoCs may be expensive. Our Composable methodology late-binds NoC decisions related to traffic to/from the interposer to SoC design time (as opposed to when designing the chiplet), thereby requiring the least effort and rework to deploy the chiplet in new SoC organizations.

Table I: Comparison of deadlock avoidance approaches.

	Independently Designed Chiplets	Enables Local Optimization	Global CDG Not Required	Future-proof Chiplets	HW Cost
VC-based	−	++	+	−	High
Flat NoCs	−−	−	−	−−	Low
Hierarchical	−	+	−	−−	Low
Composable (this work)	+	++	++	++	Low

Hardware Cost: Apart from VC-based designs, the other approaches modify the routing tables of the different NoC components and so the hardware overhead is minimal. For the VC-based approach, especially if over-provisioning for future systems is required, the area impact of supporting a larger number of VCs can become relatively costly.

IV. Multi-chiplet Routing

In this section, we present our composable, topology-agnostic, deadlock-free routing methodology for chiplet-based systems. The key insight is simple-but-powerful: from the perspective of any individual chiplet, the rest of the system (independent of the total number of other chiplets or interposer complexity) can all be abstracted away into a single virtual node, which enables tractable analysis, optimization, and correctness at a chiplet granularity. We detail one concrete approach for chiplet-based routing, but this is one possible solution that our key insight enables.

A. Overview

Before describing our methodology, we define some terms.

Definition 1 A *boundary router* of a chiplet connects the chiplet to the interposer through a *boundary link*. Traffic traversing from the interposer to the chiplet is called *inbound traffic*; traffic from the chiplet to the interposer is called *outbound traffic*.

Definition 2 The *inbound reachability* of a boundary router b, $InR(b)$, is the fraction of on-chiplet routers that can be reached from the interposer through router b; $0 < InR(b) \leq 1$.

Definition 3 The *outbound reachability* of a boundary router b, $OutR(b)$, is the fraction of on-chiplet routers that can reach the interposer through router b; $0 < OutR(b) \leq 1$.

Definition 4 The *inbound distance* of a chiplet router r, $InD(r)$, is the topological distance from the nearest boundary router that can reach r to router r.

Definition 5 The *outbound distance* of a chiplet router r, $OutD(r)$, is the topological distance from r to its nearest reachable boundary router.

The goal of the composable routing methodology is to isolate the design of individual chiplets and the interposer

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as much as possible, allowing independent load balancing optimizations on each chiplet and the interposer, while still providing deadlock-free routing for the entire system. In particular, we place unidirectional turn restrictions at boundary routers on each chiplet. When applying turn restrictions, the rest of the system is abstracted away with a single node that is connected to all boundary routers. Turn restrictions determine the inbound and outbound reachability of each boundary router and guarantee that cyclic channel dependencies do not exist within each chiplet. Then, the reachability information is propagated to the interposer, which is responsible for routing a message from one boundary router to another. With the knowledge of the boundary routers' reachabilities, messages are forwarded to the correct destination chiplet. Once a message reaches a destination boundary router, the local chiplet NoC will route the message to its final destination. This hierarchical approach uses two sets of routing tables for each chiplet. The first set of tables is used for routing messages locally within the same chiplet (this is the conventional intra-chiplet routing), while the second set steers outbound messages to the appropriate boundary routers. More implementation details are provided at the end of this section. Routing decisions corresponding to the first routing table (intra-chiplet) can be made completely independently from the rest of the system, which might not even yet be defined.

B. Chiplet Design Guidelines

When designing the chiplet-level NoC, the number and placement of boundary routers are two critical design parameters that can impact the overall system performance. These relate to the number of vertical (micro-bump) links between the chiplet and the interposer.

Number of Boundary Routers: The number of boundary routers determines the throughput a chiplet can sustain for sending/receiving off-chip traffic; the more boundary routers, the higher the off-chip traffic bandwidth. An extreme case would be to connect each router on the chiplet to the interposer with a vertical link, making every router a boundary router, as previously considered by others [2], [22]. However, such a design is likely to be over-provisioned for the expected amount of off-chip traffic and could be constrained by the available micro-bump density.

In determining the number of boundary routers per chiplet, a key observation is that while the maximum number of boundary routers possible is a function of the chiplet area, the maximum useful bandwidth is a function of its perimeter. For a chiplet with an n-by-n mesh, we have analytically determined that with the interposer topologies considered in this paper, n boundary routers are sufficient (the full analysis is omitted for brevity). For the 4×4 chiplets assumed in most of our experiments, we use four boundary routers per chiplet. While we focus on meshes, our methodology applies to other topologies (see Section VI).

Turn Restrictions at Boundary Routers: The simple example from Fig. 3 shows that there can be numerous potential dependency loops through the interposer, other chiplets, etc., leading to an explosion in the number of possible paths to be analyzed. To enable individual chiplet-level routing decisions and make the inter-chip dependency analysis tractable, we abstract away the rest of the system as a single node and connect all boundary routers with the abstract node (Fig. 4). *Unlike prior works, this novel abstraction step is the key to enabling the independent design of chiplets without requiring global CDG information.*

We use turn restrictions to break cycles containing the abstract node and a pair of boundary routers. The abstract node represents the rest of the system that designers of individual chiplets do not need to have knowledge of, hence turn restrictions do not apply to the abstract node. When choosing prohibited turns for boundary routers, connectivity must be preserved (i.e., a path must exist from each chiplet router to the abstract node, and vice versa), so turn restrictions that cause a disconnected NoC are prohibited.

Breaking all cycles while maintaining connectivity is sufficient to ensure the correctness of operation with respect to this chiplet. However while sufficient, careful selection of turn restrictions and routing are still desired for performance reasons. Different heuristics can be employed; we describe one possible approach that works well in practice. We consider inbound and outbound reachability for load balancing. An imbalanced inbound or outbound reachability can cause the chiplet and/or the interposer to become congested. Meanwhile, the average of inbound and outbound distances across all chiplet routers should be minimized, due to the fact that when routing off-chip, the nearest boundary router is preferred if a message has multiple boundary router candidates. Overall when choosing prohibited turns, our objective is to minimize $\frac{Average_distance}{Average_reachability}$, in which distance and reachability are defined in Section IV-A, and the averages are computed across all on-chip routers. To be specific, $Average_distance$ is the average of inbound and outbound distance for all routers on the chiplet. $Average_reachability$ is similarly computed over each boundary router's inbound and outbound reachability. Our heuristic selects combinations with lower average distances and higher average reachability.

To visualize the metrics, Fig. 4 gives an example for a 4×4 mesh with 3 boundary routers a, b, and c; and the rest of the system is denoted by x. Assume X-Y routing for the local chiplet NoC, and prohibited turns are denoted by the crossed-out arrows at the boundary routers. Loops containing x and any pair of boundary routers are broken by prohibiting certain turns. In this example, a has an inbound reachability ($InR(a)$) of $\frac{1}{2}$ because its inbound turn restriction, combined with the chiplet's local X-Y routing, makes the left half of the chiplet unreachable from node x through a. As there is no outbound turn restriction on a, its outbound reachability $OutR(a)$ is 1, which means every router can reach x through a. The loop of $x \rightarrow a \rightarrow c \rightarrow x$ is broken at c with an outbound turn restriction, resulting in $OutR(c) = \frac{1}{2}$. Alternatively, instead of breaking the loop at router c, an inbound turn restriction $x \rightarrow a \rightarrow (3,3)$ can be placed at router a to break the same loop, for which the $InR(a)$

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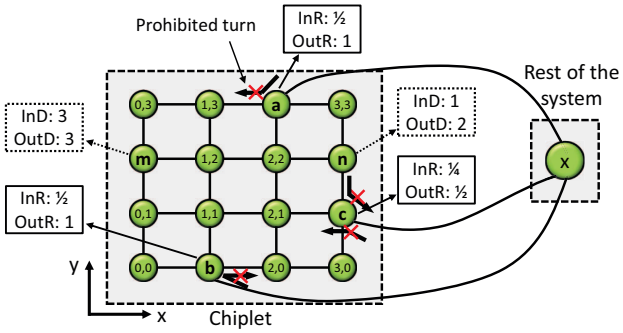

Figure 4: The effects of turn restrictions on inbound/outbound reachability of three boundary routers (a, b, and c); and on inbound/outbound distance of two routers (m and n).

becomes $\frac{1}{4}$ (only column number 2 is reachable over router a because of two separate inbound turn restrictions for breaking two different loops), and $OutR(c)$ becomes 1. Router m has an inbound distance of 3, measured from boundary router b; and an outbound distance of 3, to either a or b. Similarly, router $InD(n) = 1$ measured from router c; and $OutD(n) = 2$ to router a. While this example is somewhat ad hoc for illustrative purposes, we provide a concrete algorithm for determining all of this below.

Boundary Router Placement: Given an internal chiplet-level routing algorithm, the selection of boundary routers affects their inbound and outbound reachability and the on-chip traffic distribution. We propose the following guidelines for selecting boundary routers. First, avoid clustering boundary routers together to reduce the chance of creating network hotspots. Second, boundary routers should be placed in a way that inbound/outbound reachability for all boundary routers can be balanced. Third, routers with lower radix are preferred. The first two guidelines aim to optimize network performance and throughput. The third guideline is proposed to minimize circuit complexity. For example, in Fig. 4, the four routers in the middle of the chiplet have five ports each (one each to the neighboring routers, and a fifth link to the network endpoint it is connected to, for example, a GPU CU). Adding a vertical link to one of these "internal" routers would force the router to implement six ports, which adds area and can impact circuit timing. However, adding a vertical link to any of the routers on the periphery of this chiplet allows all routers to continue having five ports or less.

Boundary Router Placement and Turn Restriction Algorithm: Alg. 1 determines boundary router placements and turn restrictions for each chiplet. $PlaceBoundaryRouter$ iterates through all boundary router placements to find better placements and turn restrictions by calling $SetTurns$. For each placement, the function $identifyAllBoundaryTurns$ enumerates all possible boundary turns and stores them in a list $b_turn[]$. Another list $p_turn[]$ stores the prohibited boundary turns, which is updated recursively in $SetTurns$. The variable max is the maximum number of prohibited boundary turns required to eliminate all deadlocks.

Procedure $SetTurns$ examines all boundary turn combinations with heuristics, and updates the best placement found so far if current restrictions improve the user-specified objective function. We use a matrix representation of the CDG [39], [45]. Initially, all boundary turns are allowed. The $update$ function (line 16) updates the CDG with a list of prohibited boundary turns $pt[]$ that is propagated to the channel connections of the entire graph using the Floyd-Warshall all-pairs shortest path algorithm [46]. This provides the connectivity information from the updated CDG, the boundary router reachability, and the hop count. The next step checks if the user-specified objective function has been improved (line 17). In line 19, the $connected$ function checks if the CDG is still connected, as any turn restrictions that cause a disconnected network should be discarded. If the graph is connected, $hasLoop$ detects if an inbound channel (from the abstract node to a boundary router) is connected to an outbound channel (from a boundary router to the abstract node). If no loop is found, then the best placement is updated in line 22, and the recursion terminates. Line 24 controls the depth of the recursion, as only a certain number of boundary restrictions are needed to eliminate deadlock. If the CDG is connected but loops still exist, lines 27-29 invoke the recursive call to $SetTurns$ to add more turn restrictions as needed.

Algorithm 1 Set prohibited boundary turns

```
1:  procedure PLACEBOUNDARYROUTER()
2:  var
3:      b_turn[] : a list of candidate boundary turns
4:      p_turn[] : a list of prohibited boundary turns
5:      max : maximum number of prohibited turns
6:  begin
7:      for all boundary router placement do
8:          b_turn[] ← identifyAllBoundaryTurns()
9:          p_turn[] ← ∅
10:         SetTurns(b_turn[], p_turn[], 0, b_turn.size() − 1, 0, max)
11:
12: procedure SETTURNS(bt[], pt[], start, end, index, r)
13: var
14:     cdg : channel dependency graph for the chiplet
15: begin
16:     cdg.update(pt[])
17:     if objectiveFunction(cdg) not minimal then
18:         return
19:     if ! cdg.connected() then
20:         return
21:     else if ! cdg.hasLoop() then
22:         update optimal placement
23:         return
24:     else if index = r || start > end then
25:         return
26:
27:     for i = start to end do
28:         pt[index] ← bt[i]
29:         SetTurns(bt[], pt[], i + 1, end, index + 1, r)
```

C. Interposer NoC Configuration

Having determined the turn restrictions into/out of the chiplets, we now explain how to program the interposer's routing tables. Notice that the interposer network itself should also be deadlock-free when considered in isolation (without chiplets). The interposer is responsible for routing a message from one boundary router to another. To do that, certain chiplet-level information must be provided to the interposer. First, we need to know the on-chip nodes (endpoints) that are

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reachable from each individual boundary router given the turn restrictions. We use this to ensure that a message is routed to a chiplet's boundary router from which the destination can be reached. Second, we optionally use the topological distances between each boundary router and its reachable on-chip nodes to optimize routing distances and load balancing. Note that this information can simply be enumerated in a "list" format (e.g., node x is reachable from boundary node y); the full details of the chiplet's local NoC are *not* required (e.g., the topology of the network and routing decisions for how a request gets from y to x) and this information is independent of the interposer and any other chiplets.

We now describe our interposer routing scheme. For each message destined to a router on a chiplet, the following algorithm decides which boundary router of the destination chiplet to send this message to. If a destination is only reachable through a single boundary router, then the interposer must route the message to that specific boundary router. Otherwise, we pick boundary routers to balance network load across the boundary routers (equally utilizing chiplet-interposer bandwidth) while minimizing path lengths (avoid sending messages on highly-circuitous just for load balancing). Below, we formally specify the algorithm.

- For a given boundary router i, the set of nodes that are reachable ONLY by i is denoted as A_i.
- For the remaining nodes that can be reached by more than one boundary router, the list C_i contains all nodes that are topologically closer to i than any other boundary routers. For different boundary routers j and k, $C_j \cap C_k = \emptyset$.
- The remaining nodes are equidistant to at least two boundary routers. Let $E_{i,j}$ be the list of nodes that are equidistant to both boundary routers i and j. While being equidistant to more than two boundary routers is possible, we only consider the two-router case for simplicity.
- Perform the following steps to assign on-chip nodes to boundary routers.

  Step 1. Across all boundary routers, select a router i that has the minimum number of items in A_i.

  Step 2. Assign nodes from C_i to A_i one by one, until the number of items in A_i is not the smallest. An item is removed from C_i when assigned to A_i. If A_i still has the minimum number of items, assign node from $E_{i,j}$ to A_i one by one. Items are removed from $E_{i,j}$ and $E_{j,i}$ after assignment to A_i.

  Step 3. Node assignment to boundary router i is finished if no further assignment can be made. Repeat Steps 1-3, until $C_i = \emptyset$ and $E_{i,j} = \emptyset$ for all boundary routers i and j.

When finished, the node assignment information for each boundary router is stored in A_i. By referring to this information, the interposer routing table is configured accordingly. The system integrator is free to choose any underlying routing algorithm that is deadlock-free for the interposer network.

Consider the example shown in Fig. 4. For boundary routers a, b, and c, $A_a = \{(2,0),(2,1),(2,2),a\}$, $A_b = \{(0,0),(0,1),m,(0,3),b,(1,1),(1,2),(1,3)\}$, and $A_c = \emptyset$; and $C_a = \{(3,3)\}$, $C_b = \emptyset$, $C_c = \{(3,0),c,n\}$. There is no equidistant set in this network. Node assignment starts with boundary router c because A_c is empty. All the elements in C_c are assigned to A_c and $A_c = \{(3,0),c,n\}$. No further assignment can be performed for c, therefore the algorithm chooses the next router, which is a. The only element in C_a is assigned to A_a and $A_a = \{(2,0),(2,1),(2,2),a,(3,3)\}$. Up to this point, each on-chip node is assigned to exactly one boundary router; and the assignments are stored in A_a, A_b, and A_c. With the above information, the interposer is able to route a message to the correct boundary router (a, b, or c) if the message is destined to this chiplet.

D. Deadlock Freedom and Connectivity

Now we show that the composable routing scheme is deadlock free and connected. Assume that there is a cycle $r_1, l_1, r_2, l_2, ..., r_n, l_n$, in which r denotes a router and l is a link connected to r. If all of the routers and links belong to the same chiplet, then it contradicts the basic assumption that the chiplet-level network is deadlock free. Otherwise, if a subset of the cycle belongs to the interposer and other chiplets, this subset can be abstracted with a single node x. Therefore, the cycle is converted into $r_1, l_1, ..., r_j, x, r_k, l_k, ..., r_n, l_n$. Because all cyclic dependencies in loops containing x are removed, the new cycle is deadlock free. As a result, the composable routing scheme is deadlock free.

Any network within a single chiplet is connected, because boundary router turn restrictions do not affect the internal chiplet network. Any node on a chiplet is able to reach the interposer through at least one boundary router. The interposer network is connected by construction (i.e., every interposer router can reach every other interposer router). Given any pair of on-chip nodes, a path exists between the two nodes. As a result, the system is connected.

E. Microarchitectural Issues

Each chiplet needs to implement two different routing tables. The first handles intra-chiplet traffic that never goes to the interposer. This routing table may be populated in whatever manner the chiplet designer deems appropriate. The second routing table directs outbound traffic to the appropriate boundary router. This organization assumes a global ID space for all of the router endpoints throughout the collective system. Analogous to the boot-up sequences used to detect all of the memory and compute resources available in a system (especially in a multi-socket SMP system), composable interposer-based SoCs would need a similar protocol for system configuration. Part of this process would be the detection of the available NoC endpoints, assignment of unique IDs to each endpoint, and the computation and population of the secondary routing tables. Unlike system boot-up, this process would only be performed once by the SoC integrator after physically assembling the SoC (although hooks may also be provided to update the tables at a later point in time, such as to handle failed links [32]).

In our design, each network interface (NI) has a lookup table that maps the destination ID of an outbound packet to a

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boundary router ID. The boundary router ID is then embedded in the header flit and used for intra-chiplet routing until the packet leaves the chiplet. Regarding area/power overhead, the lookup table in each NI needs to provision against the largest system size for a given generation of products. Routing tables are typically much smaller than other router components such as buffers and crossbars. Furthermore, the size of the second routing table in each chiplet is only proportional to the number of boundary routers; thus, it is significantly smaller than the first routing table. There are several ways to implement the interposer router: 1) Provision the routing tables for the largest system size, resulting in relatively large interposer routing tables; or 2) Add another layer of destination mapping to convert destination IDs to destination boundary router IDs, leading to smaller routing tables but more complex boundary routers. Overall, our design should not incur significant additional power/area/timing impact compared to a canonical two-stage router.

V. Evaluation

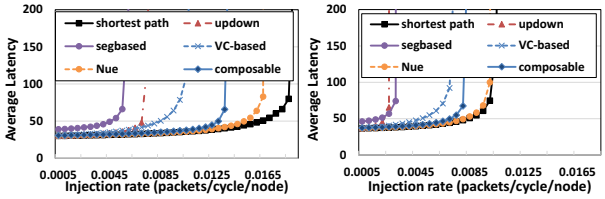
A. Experimental Methodology

To evaluate network performance, we use an APU simulation platform consisting of gem5 [47] and a modified version of the GPU model [48] for cycle-level execution-driven simulations. We use Garnet [49] to simulate the network using 2-stage routers with 4-flit buffers per channel. Our initial experiments use the multi-chiplet APU configuration shown in Fig. 2 consisting of four GPU chiplets, one CPU chiplet, and an active interposer. The CPU chiplet consists of CPU cores, private CPU L1 and L2 caches, and a last level cache. Each GPU chiplet consists of 16 compute units (CUs) and 8 GPU L2 cache banks. Our memory model utilizes the built-in gem5 model [50] with eight memory channels and eight banks per channel. Fig. 2 also shows the placement of the boundary nodes as determined by our algorithm from Section IV.

We use both synthetic traffic and application-based simulation. For synthetic traffic, each packet is 8 flits wide, and the network is simulated for 2 million cycles. For system-level (non-synthetic) simulations, we use APU applications from the AMD SDK [51], Rodinia [52], and Pannotia [53] suites, where off-chiplet communication includes both cache coherence between GPU CUs and traffic to main memory.

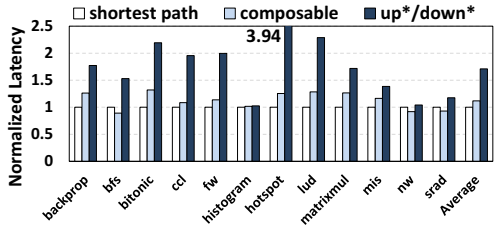
B. Comparison Points

Even though qualitatively the VC-based approach is expensive and unattractive, we provide a comparison for completeness. Using a similar methodology as EbDa [40], we implement a VC-based deadlock avoidance mechanism that supports minimal-path adaptive routing using four VCs: two VCs are needed to avoid deadlock for a single 2D mesh; by introducing vertical connections between chiplet and the interposer, two more VCs are required to isolate inbound and outbound traffic. We also implement three global routing algorithms described in Section III-C: up*/down* [37], segment-based [38], and Nue routing [39]. Note that all three require full CDG knowledge and do not enable independent

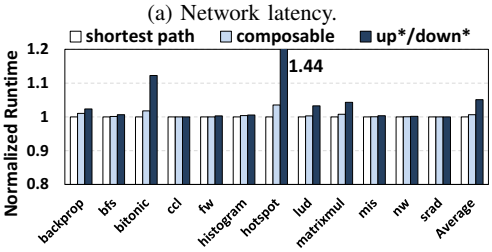


(a) Uniform random. (b) Bit complement.

Figure 5: Load-latency curve w/ synthetic traffic.



(a) Network latency.



(b) Execution time.

Figure 6: System performance under realistic traffic.

design and chiplet reuse. We compare against them as the most relevant work that we are aware of, but they fail to satisfy our key chiplet modularity criteria.

The root node in up*/down* routing is selected by finding the node with the lowest average distance to all other nodes before applying turn restrictions. The starting segment in segment-based routing is formed from the top-left of the system (the top-left router of the top-left GPU chiplet). In our composable routing scheme, both the local chiplet algorithms and the interposer algorithm use dimension-ordered routing. For fair comparison against the VC-based approach, we provide four VCs for each of the turn-based schemes.

We also compare our results to an idealized system (denoted as **shortest path**) that uses an impractically large number of virtual channels to avoid deadlock. The routing tables are configured using an all-pairs shortest path (APSP) algorithm (in contrast to the prior-art and our own proposed scheme where some routes may not be minimal). Note that this idealized system does not necessarily provide true optimal performance because APSP can still lead to higher levels of congestion in some links compared to others. However in practice, we find that this shortest-path configuration typically outperforms the practical alternatives, and so it provides an optimistic performance target to compare against.

C. Basic Throughput Evaluations with Synthetic Traffic

In this section, we evaluate a 64-CU system consisting of four chiplets, each with 16 CUs organized as a 4×4 mesh.

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Each chiplet is connected to the interposer through four boundary routers. The interposer network is a 4×4 mesh.

Fig. 5a and Fig. 5b show the load-latency curves under uniform random and bit complement traffic. We observed that many heterogeneous multi-chiplet workloads have similarities to uniform random traffic: the real system has a mix of intra-chiplet, inter-chiplet, chiplet-to-interposer, and chiplet-to-memory traffic covering both coherence and main memory requests and responses; these in aggregate "average out" such that the high-level performance trends of uniform random traffic largely match several of our application-driven studies. The bit complement traffic pattern forces all packets to go off-chiplet, therefore it further stresses the interposer and creates network hotspots. We ran other synthetic traffic patterns, but the overall trends were very similar so they were not shown.

Our composable scheme outperforms up*/down*, segment-based, and VC-based approaches.[1] With the same number of VCs, the composable scheme performs better than the VC-based approach mainly because the extra VCs reduce head-of-line blocking. Typical coherence protocols require between 3-5 virtual networks, each of which would require four VCs for deadlock freedom in the VC-based approach. A coherence protocol for a heterogeneous architecture may well require more virtual networks making the VC-based approach even more expensive. While the VC-based approach requires four VCs/virtual network for correctness, more VCs are needed for performance as indicated by the performance gap between the VC-based and composable schemes.

Segment-based routing suffers from larger zero-load latency and has the lowest saturation throughput. This is mainly because it is designed and optimized for 2D mesh-like networks. Although the evaluated system consists of multiple mesh networks, the global topology remains irregular such that segment-based routing cannot efficiently handle it. The baseline segment-based algorithm does not always form an optimal segment; a segment starting from a boundary router toward the interposer is likely to wrap around and end on a router on the same chiplet, or traverse multiple hops through another chiplet until it reaches a router belonging to an existing segment. Such chain-like segments can be very long in larger systems, and breaking any bidirectional turns within the segment will result in more non-minimal paths (for the baseline APU, we observed an average routing distance of nearly 11 hops, as opposed to ∼8 hops for the other approaches). Although topology-aware optimization might improve the performance of segment-based routing, it is out of the scope of this paper.

Up*/down* routing has low zero-load latency, which indicates that messages are likely to take minimal routes in the evaluated system. However, it saturates relatively early compared to the other approaches. Links near the root node are inherently more congested than those near the leaf nodes. When injection rate increases, these links saturate and become bottlenecks. Nue routing outperforms our composable approach, but does so only because it has the benefit of optimizing its routing with knowledge of the full CDG, resulting in similar behavior to the ideal shortest-path algorithm. With sufficient VCs (which we provision it with), it finds optimized paths to balance the network workload.

Our composable scheme outperforms up*/down* and segment-based routing because the chiplet and interposer networks are more load balanced, as are the vertical links between the chiplets and the interposer. Nue provides better load balancing and therefore performs close to idealized shortest-path routing, but like the other prior works it is not applicable for independent design and reuse of chiplets for modular SoC construction. Compared to the idealized shortest-path routing, our scheme covers much of the throughput gap from up*/down*, but there remains some headroom. This is because of 1) some remaining load imbalance due to turn restrictions, and 2) the idealized network has more virtual channels to improve head-of-line blocking. Overall, although our proposed approach does not achieve the full performance of globally-load balanced optimization, our results show that our scheme ensures correctness while delivering high performance for multi-chiplet SoCs, and it uniquely enables a modular chiplet-based design methodology that does not require *a priori* knowledge of the full system's CDG.

D. Application-level Impact

Network Latency: We evaluate our composable routing scheme with non-synthetic workloads using execution-driven simulation. Fig. 6a shows the average network latency, normalized to the idealized shortest-path approach. Segment-based routing is not shown because it is consistently and significantly out-performed by the remaining approaches at the given system size. We omit the VC-based scheme because the evaluated heterogeneous system requires an impractically large number of VCs to avoid routing and protocol-level deadlock while retaining performance. Overall, our composable approach achieves network latencies that are nearly the same as the shortest-path case. There are a few cases (bfs, nw, srad) where the composable routing performs marginally better than shortest path; as discussed earlier, shortest-path is not truly optimal, and there are occasionally situations where localized bursts of traffic (which occurs more often in GPU workloads than conventional CPU applications) can cause congestion/load imbalance in the shortest-path configuration.

The up*/down* approach suffers over 50% increases in average network latency for most benchmarks because the root node becomes the bottleneck under heavy traffic, as discussed in Section V-C. Such bottlenecks limit the system's effective bandwidth and lead to significant in-network buffering delays.

Application Performance: Fig. 6b shows the program execution time, normalized to the idealized **shortest path** approach. Overall, composable routing achieves similar (within 1%) system performance compared to shortest path. While APU/GPGPU applications generate large bursts of

[1] In the absence of network contention, head-of-line blocking, etc., the theoretical saturation throughput for uniform random and bit complement traffic are 0.031 and 0.016 packets/cycle/node, respectively.

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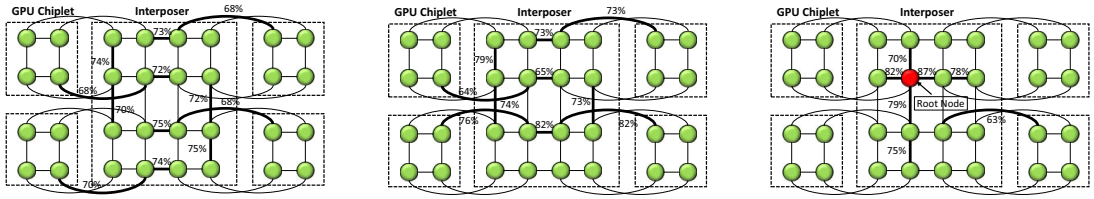


(a) Shortest path routing. (b) Composable routing. (c) Up*/down* routing.

Figure 7: Maximum link utilization for HotSpot application.

NoC/memory traffic (which is great for stressing the network), the overall impact on application execution time is muted because most GPU applications are inherently less sensitive to latency (i.e., increased latency can be more easily tolerated by abundant SIMD parallelism). While there are bursts of traffic, significant portions of the applications do not operate the NoC near saturation, and so there is less impact on total execution time. Nevertheless, we still observe a 5-10% performance degradation with up*/down* routing for some workloads while our approach performs at about the same level as the shortest-path approach.

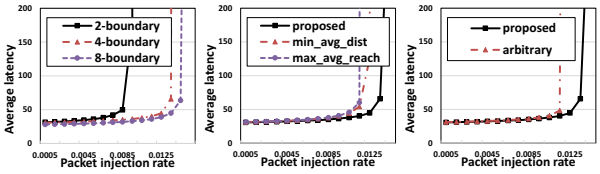
Case Study – HotSpot: Fig. 7 shows the maximum link utilization for the most heavily-utilized links when executing HotSpot. Only the interposer network and boundary routers on the GPU chiplets are shown, as the rest have low utilization. For every 10000-cycle period, we sampled the utilization of each link. The maximum utilization for a link is the largest sampled result observed over the course of the entire program execution. Maximum link utilization shows us where the worst link congestion occurs under bursty traffic behavior, which in turn allows us to visualize the global network traffic flows and locate any NoC bottlenecks. In general, composable routing has fewer congested links than shortest path. However, the former has a slightly unbalanced traffic distribution on the interposer as indicated by the larger maximum link utilization. This is caused by turn restrictions that bias the reachability of boundary routers (i.e., some boundary routers receive more traffic). For up*/down* routing, the root node resides on the interposer. As expected, links near the root node are much more utilized than the others.

VI. Broader Applicability

The previous section demonstrated the effectiveness of our methodology for one specific chiplet-based SoC. In this section, we provide additional experimental results as a variety of system assumptions are modified, and then we also discuss how the proposal can be applied to chiplet-based systems *without* active interposers.

A. Design Guideline Justification

In Section IV-B, we described how to determine the number of boundary routers, the objective function to select turn restrictions, and the boundary router placement. To demonstrate the effectiveness of the proposed guidelines, we evaluate additional design alternatives with uniform random traffic. Fig. 8a shows the throughput improvement when increasing the number of boundary routers from 2 to 8. In



(a) Number of boundary routers (b) Objective function (c) Boundary routers placement

Figure 8: System performance with various design alternatives under uniform random traffic.

all cases, the interposer network remains the same size; with 8 boundary routers, 2 boundary routers are concentrated to one interposer router, which increases router complexity and area. The improvement from 4 to 8 boundary routers is much smaller than that from 2 to 4 routers. Insufficient boundary routers can impact system throughput. By providing more boundary routers, off-chiplet bandwidth is increased, which reduces interference with intra-chiplet communication. In an extreme case of 16 boundary routers, where each chiplet router has a vertical connection, there will be no intra-chiplet routing/congestion with off-chiplet traffic. However, such a design is impractical due to the large number of vertical wires. Overall, 4 boundary routers is a reasonable design choice in terms of performance and hardware cost.

Fig. 8b compares the system throughput for different objective functions, including minimizing average distance, maximizing average reachability, and our proposed metric (i.e., minimizing $\frac{Average_distance}{Average_reachability}$). Results show that our proposed objective function is effective and provides the best performance compared to the other metrics. Considering only average distance or average reachability tends to create unbalanced on-chiplet traffic.

In some situations, the designer may not have the freedom to choose where the boundary routers are placed (e.g., layout restrictions, physical design constraints). Fig. 8c considers a configuration where the boundary router locations have been moved to less-optimal locations (e.g., clustered together), spread to corners, on the same row (i.e., different from where our approach would assign them to). We re-ran our algorithm for determining turn restrictions. Results show that the random placement of boundary routers ends up causing some links to be used more than others, therefore impacting system throughput, but deadlock freedom is maintained.

B. Sensitivity Studies

We consider the following variants of our baseline:

System Size: The baseline has 4 GPU chiplets with 16 CUs each, for a total compute capacity of 64 CUs. We

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also consider two 128-CU configurations (CPU count held constant) consisting of (1) 4 chiplets with 32 CUs each, and (2) 8 chiplets with 16 CUs each. In both cases, there are still four boundary routers per chiplet.

Interposer NoC Topology: To support the claim that the interposer's NoC can be designed independently from the chiplets, we evaluate our baseline system but replace the interposer's mesh NoC with a "Double Butterfly" topology [22].

Irregular Chiplet Topologies: To support the similar claim that the chiplets' NoC topologies can be independently designed, we evaluate a system where each GPU chiplet implements a different local NoC topology consisting of mesh, ring, butterfly, and tree topologies.

Results: The analyses in this section are presented only with the load-latency curves for uniform random traffic. We did perform application-level experiments as well, but the overall trends are very consistent, and so we omit those additional graphs due to space reasons and their repetitiveness. The main point of these experiments is to demonstrate that our proposal is a robust way to achieve high performance while ensuring deadlock freedom across a wide range of chiplet-based system possibilities.

Fig. 9a and b show the results for the larger 128-CU configurations, with our composable approach handily outperforming up*/down* routing. The major difference between these two configurations is the ratio of intra-chiplet to inter-chiplet traffic. Compared to shortest path, our composable approach is less sensitive to traffic distribution due to better chiplet-level and interposer-level load balancing.

Fig. 9c shows the results when the interposer NoC has a butterfly-based topology. The results are similar to the baseline system with the mesh, and overall this helps to demonstrate that the individual chiplets can be easily designed independent of the interposer's NoC topology.

Fig. 9d shows the results when each of the GPU chiplets has a different local NoC topology. The results here are more interesting as our proposal results in a higher saturation throughput than the "ideal" case with copious virtual channels and shortest path routing. This is because when handling inter-chiplet communication, shortest path favors boundary routers near the center of the interposer while our proposed approach achieves better distribution of interposer traffic.

C. Other Chiplet Packaging Options

Thus far, our studies have focused on chiplet-based systems built on emerging active silicon interposer technologies. While active silicon interposers can be practical, especially if the total amount of interposer area used for logic can be minimized [2], [30], near-term chiplet systems may be limited to passive substrates. Whether using a passive silicon interposer [19]–[21] or a more conventional packaging substrate [7], [11], [14], [54], one possible concern is that the lack of an active layer below the chiplets could constrain the applicability of our methodology.

Fig. 10a shows an example system with chiplets on a passive substrate. This layout assumes a central chip that

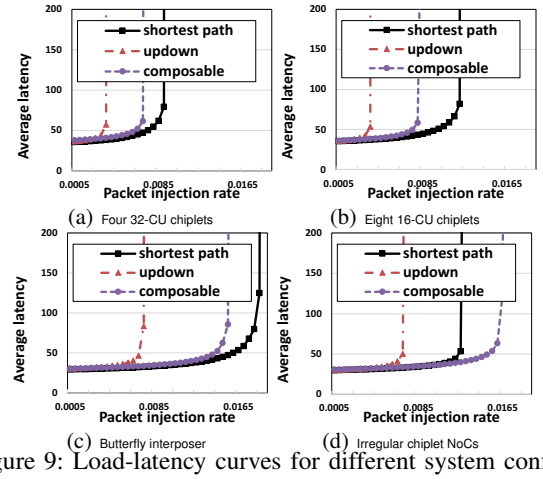

Figure 9: Load-latency curves for different system configurations under uniform random traffic.

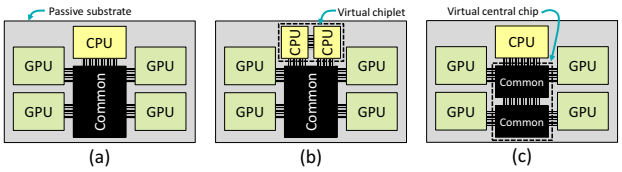

Figure 10: Chiplet-based systems on passive substrates (e.g., passive interposer, MCM).

provides the common functionality that would otherwise be put on the active interposer (e.g., memory controllers, NoC, system management), with the compute chiplets fanning out from the central chip in a star-like topology. With this type of layout, our proposed methodology can be applied directly to this system without any modification by treating the central chip in the same way as the active interposer in our previous working example. The process for selecting the best boundary node placement can be more efficient as there are fewer reasonable nodes on a chiplet to choose from (i.e., those closest to the central chip).

D. Other Chiplet Topologies

Even for chiplets in a non-star topology, our approach can be adapted to work. Fig. 10b shows a chiplet-based system where two CPU chiplets have additional point-to-point links (e.g., for low-latency cache coherence) that do not route through the central chip. To support this, the two CPU chiplets are effectively treated as a single virtual chiplet in order to apply our methodology to determine routing restrictions. It is still up to the CPU chiplet designer to ensure that the routing directly between the two CPU chiplets is correct (i.e., deadlock free), but the designers need not worry about traffic entering/leaving *either* of the chiplets from/to the central chip as our methodology determines the appropriate turn restrictions to ensure proper operation of the overall SoC.

Likewise, Fig. 10c shows a system where there is no single "central" chip, but rather there are two chips that the other chiplets connect to. Here, we apply a similar technique where the two chips are treated as a single virtual chip for the purposes of this approach. Similar to the two-CPU chiplet example above, the SoC designer must ensure that

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the pair of chips are mutually/locally deadlock free, but any remaining connections to the other chiplets will be correctly taken care of. Most reasonable chiplet topologies can be iteratively coalesced until the topology is converted into a star-like organization, at which point our approach can be directly applied.

VII. Related Work

Flat Networks: In Section III-C, we introduce a flat-network approach to deadlock avoidance. Further optimizations to up*/down* routing have been proposed: Koibuchi *et al.* construct a left-to-right directed graph based on a BFS spanning tree and distribute the traffic around the root node [55]; Sancho *et al.* use a depth-first search (DFS) spanning tree [56]; they improve traffic balance by removing channel dependencies separately for each direction in each cycle [57].

Hierarchical Networks: HiRA [41] is a methodology for deadlock free routing in hierarchical NoCs. In HiRA, the network is divided into subnets (networks with independent deadlock-free routing algorithms), and external links (links between subnets). Deadlock is avoided by selecting safe boundary nodes in each subnet and applying turn restrictions on boundary nodes. When connected to other subnets, a boundary node is safe if a deadlock cannot occur and connectivity is guaranteed without modifying the subnet's internal routing algorithm. A CDG consisting of all boundary nodes is used when applying turn restrictions on boundary nodes. While HiRA can be applied to a chiplet-based system with a passive interposer, it is not applicable to active-interposer SoCs for two major reasons. First, a system-level CDG is still required and turn restrictions are largely dependent on subnet routing algorithms. Second, HiRA lacks a routing algorithm for the central network (i.e., the active interposer), which is connected to all chiplets.

Routing in 3D NoCs: Deadlock avoidance techniques for regular 3D NoCs include DoR and turn-based routing [35], [58]–[60], and VC-based approaches [61]–[63]. Many of the techniques are not directly applicable because their turn-based algorithms require that every router has vertical connectivity to other layers in the stack (we do not make this assumption), which increases per-chip TSV area overheads. Other 3D VC techniques create monotonic VC orderings tied to the chip's vertical position in a stack; the physical topology of chiplets on an interposer makes it difficult to impose a total ordering. We do not provide experimental evaluations against these works as it is not obvious how to adapt them to topologies not consisting of a single, vertical, 3D chip stack.

VIII. Conclusions

While the of chiplet-based construction of complex SoCs is very exciting for the types of systems that it can enable, such systems must be *easy* to design and assemble. When the system is constructed using what may become black-box chiplets from third-party silicon IP vendors, ensuring correctness becomes even more challenging and important. This paper makes a significant contribution toward chiplet-based SoC design methodologies focused on the interconnect; however, there remain other fruitful areas for research. If the different chips in the system are to be cache coherent, then one must devise a cache coherence protocol that operates correctly and scales out (in terms of performance) across the disparate physical chips. While not a strict correctness issue, it is likely that quality-of-service mechanisms need to be devised to ensure that the different chips integrated together "play nicely" with each other, especially with real-time components (e.g., graphics and audio) or higher-level performance targets (e.g., datacenter service-level agreements).

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